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Multinomial Distribution

[< Previous](#)[Next >](#)

Multinomial Distribution

Multinomial distribution is a generalization of binomial distribution.

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[NumPy Array Split](#)
[NumPy Array Search](#)
[NumPy Array Sort](#)
[NumPy Array Filter](#)

NumPy Random

[Random Intro](#)
[Data Distribution](#)
[Random Permutation](#)
[Seaborn Module](#)
[Normal Distribution](#)
[Binomial Distribution](#)
[Poisson Distribution](#)
[Uniform Distribution](#)
[Logistic Distribution](#)
[Multinomial Distribution](#)
[Exponential Distribution](#)
[Chi Square Distribution](#)
[Rayleigh Distribution](#)
[Pareto Distribution](#)
[Zipf Distribution](#)

NumPy ufunc

[ufunc Intro](#)

It has three parameters:

n - number of times to run the experiment.

pvals - list of probabilities of outcomes (e.g. [1/6, 1/6, 1/6, 1/6, 1/6, 1/6] for dice roll).

size - The shape of the returned array.

Example

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Draw out a sample for dice roll:

```
from numpy import random

x = random.multinomial(n=6, pvals=[1/6, 1/6, 1/6,
1/6, 1/6, 1/6])

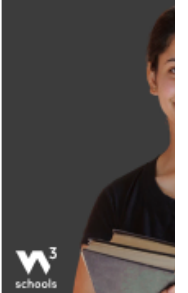
print(x)
```

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Note: Multinomial samples will NOT produce a single value! They will

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ufunc Summations
ufunc Products
ufunc Differences
ufunc Finding LCM

produce one value for each `pval`.

Note: As they are generalization of binomial distribution their visual representation and similarity of normal distribution is same as that of multiple binomial distributions.

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Exercise ?

The `random.multinomial()` method has three parameters, which ones?

- ☐ `loc scale size`
- ☐ `n pvals size`
- ☐ `loc dev size`

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